

Weak Piola identity for a compactly supported test field

MathlibAnnex.BilipschitzOrientation.weak_piola

theorem

Uses coordinate perturbations to show that the signed pullback of a test-field divergence has zero integral.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

Write $Df(x) : E \rightarrow E$ for the Fréchet derivative of f at x when it exists, and set $Df(x) = 0$ at other points. Define the signed Jacobian by $J_f(x) = \det Df(x)$.

Let $W : E \rightarrow E$ be a C^1 vector field, and choose a compact set $K \subseteq T$ such that $W(y) = 0$ for $y \notin K$. Write $W_i(y) = (W(y))_i$ and define

$$\operatorname{div} W(y) = \operatorname{tr} DW(y) = \sum_{i=1}^n \partial_i W_i(y).$$

All integrals use Lebesgue measure λ on these fixed coordinates. Then

$$\int_S (\operatorname{div} W)(f(x)) J_f(x) d\lambda(x) = 0.$$

Assumptions

The field W is continuously differentiable on all of E . Its chosen compact set K lies in T and contains $\{y : W(y) \neq 0\}$; K need not equal the closure of that set. No second derivative of f is assumed.

The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

The integral of the signed Jacobian times the pulled-back divergence vanishes. For $n = 0$, the divergence is an empty sum and the conclusion is immediate. In positive dimension the proof below integrates determinant differences produced by compactly supported perturbations.

The argument integrates each determinant difference as one integrable function. It never subtracts two possibly divergent integrals over the whole ambient space. The absolute values in the integrability estimates do not replace the signed determinant in the theorem.

Proof route

For each coordinate, construct a compactly supported perturbation u_i of f . Its determinant difference equals one term of the desired integrand. Prove that difference is integrable, use the null-Lagrangian identity to make its integral zero, and sum the coordinate identities.

▼ Proof steps

1. Obtain a compact preimage and a Lipschitz test field. The relevant identities are

$$W(y) = 0, \quad DW(y) = 0 \quad (y \notin K),$$

$$f^{-1}(K) = g(K) \subseteq S, \quad g(K) \text{ is compact.}$$

The derivative vanishes off K because the complement of the closed set K is open and W is identically zero there. Continuity of DW on the compact set K gives a number $M \geq 0$ such that $\|DW(y)\| \leq M$ for every $y \in E$. The compact-field Lipschitz lemma applies to this global derivative bound and the C^1 field W , and yields a global Lipschitz bound for W .

For the preimage identity, the lower metric estimate makes f globally injective. If $f(x) = y \in K \subseteq T$, then $f(g(y)) = y = f(x)$, so $x = g(y)$. Conversely $f(g(y)) = y$ for $y \in K$. This is the exact preimage formula. The set $g(K)$ is compact because g is continuous.

2. Define the coordinate perturbations. The case $n = 0$ is already settled; write $n = m + 1 > 0$. For $1 \leq i \leq n$, let e_i be the i th coordinate vector and let $P_i z = z_i e_i$. Define

$$u_i(x) = P_i(W(f(x))) = W_i(f(x))e_i, \quad F_i(x) = f(x) + u_i(x),$$

$$K_i = \overline{\{x : u_i(x) \neq 0\}} \subseteq f^{-1}(K) = g(K).$$

The inclusion holds because $W(f(x)) = 0$ whenever $f(x) \notin K$, and $g(K)$ is closed. Hence K_i is compact. The maps P_i , W and f are Lipschitz, so u_i and F_i are globally Lipschitz. Write $DF_i(x)$ for the derivative of F_i where it exists and for the zero linear map otherwise. These are the maps and compact supports used in Step 4.

3. Calculate the determinant difference. At a point x where f is differentiable, put $A = Df(x)$ and $B = DW(f(x))$. Then

$$B_{ij} = (DW(f(x))e_j)_i = (\partial_j W_i)(f(x)), \quad DF_i(x) = (I + P_i B)A.$$

The matrix $I + P_i B$ has the identity rows except that row i is $e_i^\top + \sum_j B_{ij} e_j^\top$. Row-linearity gives

$$\det(I + P_i B) = 1 + B_{ii}.$$

Indeed, a term with $j \neq i$ has two equal rows, whereas the $j = i$ term is B_{ii} . Thus

$$\det DF_i(x) - J_f(x) = (\partial_i W_i)(f(x)) J_f(x).$$

Define, on all of E ,

$$Q_i(x) = \det DF_i(x) - J_f(x), \quad H_i(x) = (\partial_i W_i)(f(x)) J_f(x).$$

By Rademacher's theorem for the globally Lipschitz map f , the calculation proves $Q_i = H_i$ for λ -almost every $x \in E$.

4. Prove integrability and apply the null-Lagrangian identity. For the maps and compact set from Step 2,

$$\int_{K_i} |\det DF_i(x)| d\lambda(x) < \infty, \quad \int_{K_i} |J_f(x)| d\lambda(x) < \infty,$$

$$Q_i(x) = 0 \quad (x \notin K_i).$$

The two finite integrals follow from compact-set integrability of derivative minors, applied separately to F_i and f on K_i . In dimension $n = m + 1$, selecting all n rows in coordinate order makes the minor the determinant itself. Outside K_i , u_i vanishes on a neighborhood, so $F_i = f$ there and their total derivatives agree. This is the off-support determinant-difference identity. Consequently,

$$\begin{aligned} \int_E |Q_i(x)| d\lambda(x) &= \int_{K_i} |\det DF_i(x) - J_f(x)| d\lambda(x) \\ &\leq \int_{K_i} |\det DF_i(x)| d\lambda(x) + \int_{K_i} |J_f(x)| d\lambda(x) < \infty. \end{aligned}$$

The null-Lagrangian output is

$$\int_E Q_i(x) d\lambda(x) = \int_E (\det D(f + u_i)(x) - \det Df(x)) d\lambda(x) = 0.$$

Here the compact-perturbation theorem is applied to these same f, u_i : both are globally Lipschitz, u_i has compact support, $n = m + 1 > 0$, and the selected minor uses every coordinate in order. Finally, the almost-everywhere equality in Step 3 gives

$$H_i \in L^1(E, \lambda), \quad \int_E H_i(x) d\lambda(x) = \int_E Q_i(x) d\lambda(x) = 0.$$

5. Restrict to the source set and sum. For $x \notin S$, Step 1 gives $f(x) \notin K$, and therefore $DW(f(x)) = 0$ and $H_i(x) = 0$. Hence

$$0 = \int_E H_i d\lambda = \int_S H_i d\lambda + \int_{E \setminus S} H_i d\lambda = \int_S H_i d\lambda.$$

Using the defining coordinate sum for divergence, we obtain

$$\begin{aligned}\int_S (\operatorname{div} W)(f(x)) J_f(x) d\lambda(x) &= \int_S \sum_{i=1}^n H_i(x) d\lambda(x) \\ &= \sum_{i=1}^n \int_S H_i(x) d\lambda(x) = 0.\end{aligned}$$

The sum may pass through the integral because each H_i is integrable by Step 4.

Main citations

- Exact declaration and its source — `MathlibAnnex.BilipschitzOrientation.weak_piola`
- A compact C^1 field has a global Lipschitz bound —
`MathlibAnnex.BilipschitzOrientation.exists_lipschitzWith_compactField`
- Compact carrier preimages through the inverse —
`MathlibAnnex.BilipschitzOrientation.preimage_eq_g_image`
- Compact-set integrability of derivative minors —
`MathlibAnnex.NullLagrangian.integrableOn_maximalMinor_fderiv_of_lipschitzWith`
- Minor difference outside compact support —
`MathlibAnnex.NullLagrangian.topMinor_difference_eq_zero_of_not_mem_tsupport`
- Compact perturbation null-Lagrangian identity —
`MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith`
- Rademacher for globally Lipschitz maps in finite dimension — `LipschitzWith.ae_differentiableAt`

Lean source signature (exact)

```
theorem weak_piola {n : ℕ} (D : BilipschitzOpenData n)
  (W : CompactC1VectorField (Fin n → ℝ)) (hW : W.carrier ⊆ D.target) :
  ∫ x in D.source,
    divergence W (D.f x) * (fderiv ℝ D.f x).det ∂volume = 0
```

In the source	Mathematical meaning
<code>(D : BilipschitzOpenData n)</code>	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name D is not the derivative symbol.
<code>D.source; D.target; D.f; D.g</code>	The local mathematical objects S, T, f, g , respectively, throughout this Card.
<code>(W : CompactC1VectorField (Fin n → ℝ))</code>	The pair consisting of a chosen compact carrier K and a global C^1 vector field $W : E \rightarrow E$ with nonzero set contained in K , in the cited test-field type.
<code>(hW : W.carrier ⊆ D.target)</code>	The carrier satisfies $K \subseteq T$. It contains the closed support of W but need not equal it.

In the source	Mathematical meaning
<code>divergence W (D.f x)</code>	The pulled-back divergence $(\operatorname{div} W)(f(x))$, where $\operatorname{div} W = \operatorname{tr}(DW)$.

In the source	Mathematical meaning
<code>(fderiv ℝ D.f x).det</code>	The signed Jacobian $J_f(x) = \det Df(x)$.
<code>∫ x in D.source, divergence W (D.f x) * (fderiv ℝ D.f x).det ∂volume = 0</code>	The output $\int_S (\operatorname{div} W)(f(x)) J_f(x) d\lambda(x) = 0$. Compact perturbations in the proof are auxiliary constructions, not added inputs.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 19–25:

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators ENNReal NNReal Topology

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.weak_piola`
Accepted content SHA-256: `0c048dc1cede12ab8b6f85e6951993e567470888327ea3ddcbbfab5c32a128b`
Accepted source guide SHA-256: `c47a6385914233bc64b3984cfd992899a3872ac1e5205c8bce2b871a55ceffa`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`